

1. Administrative & Study Identification

Full Study Title: A Study to Investigate Safety, Tolerability, Pharmacokinetics (PK), Pharmacodynamics (PD), and Immunogenicity of R07669330 in Participants With Geographic Atrophy (GA) Secondary to Age-related Macular Degeneration (AMD) **Short Title/ Acronym:** R07669330 in GA Secondary to AMD **Protocol Number:** BP45482 **NCT Number:** NCT06961370 **Posting ID:** 602021027169402964 **Sponsor/Company:** Hoffmann-La Roche (Roche) **Investigational Product:** R07669330 **Phase of Development:** Phase 1 **Trial Status:** Recruiting (Currently recruiting participants) **Medical Monitor:** global-roche-genentech-trials@gene.com

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the ocular and systemic safety and tolerability of R07669330 in participants with Geographic Atrophy (GA) secondary to Age-related Macular Degeneration (AMD) in at least one eye in Part 1, or both eyes in Part 2, after multiple unilateral intravitreal (IVT) doses. **Secondary Objectives:** To evaluate the Pharmacokinetics (PK), Pharmacodynamics (PD), and Immunogenicity of R07669330, alongside changes in visual acuity and ocular anatomy.

2.2 Background & Rationale To evaluate the safety, tolerability, and exploratory efficacy of R07669330 as a novel treatment option to slow disease progression and prevent irreversible vision loss in patients suffering from geographic atrophy (GA) secondary to age-related macular degeneration (AMD).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Multi-part study) **Control Type:** Active-comparator-controlled (Syfovre or Izervay) in Part 2 **Blinding:** Part 1A: Open-label; Part 1B and Part 2: Masked **Randomization:** Part 1A: Non-randomized (Single-group); Part 1B and Part 2: Randomized, parallel-group

3.2 Planned Enrollment & Duration Target \$N\$: 132 participants **Number of Sites:** Multicenter **Estimated Study Start:** 16-Jul-2025 **Estimated Primary Completion:** 03-Jan-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Geographic Atrophy (GA) secondary to Age-related Macular Degeneration (AMD). 2. Clear ocular media, adequate pupillary dilation, and the ability to fixate for quality fundus imaging. 3. Best Corrected Visual Acuity (BCVA) requirements vary by part: Part 1A requires 19-48 letters read; Part 1B requires >19 letters; Part 2 requires ≥ 24 letters.

4.2 Exclusion Criteria 1. Macular atrophy in either eye due to causes other than AMD. 2. Prior treatment with any approved therapy for GA ≤ 20 weeks prior to Day 1 (in either eye for Part 1A/2; in the study eye for Part 1B). 3. Evidence of prior or active choroidal neovascularization (CNV) for Part 2. 4. Non-functioning non-study eye (defined as BCVA of hand motion or worse, and/or no physical presence of the eye).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple unilateral intravitreal (IVT) doses of R07669330 at escalating dose levels across distinct cohorts. **Route of Administration:** Intravitreal (IVT) injection. **Duration of Treatment:** Ongoing through primary completion (approximately 18 months).

5.2 Concomitant & Prohibited Therapy Permitted: Standard ophthalmological care that does not interfere with the study parameters. **Prohibited:** Prior or concurrent treatment with other approved GA therapies within 20 weeks prior to Day 1. This includes active comparators/therapies such as pegcetacoplan (Drug Preferred Name: pegcetacoplan; ATC Code: L04AA54; Trade Names: Syfovre, Empaveli) and avacincaptad pegol (Izervay).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Informed Consent, BCVA, SD-OCT, FFA, CFP imaging
2	0	Baseline
3	0	Randomization (Part 1B/2), First IVT Injection, IOP check
4	Per Protocol	Safety evaluation, systemic tolerability checks, IVT dosing
5	Jan 2027	End of Study
6		Final ocular imaging, PK/PD blood draws, Exit evaluation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence, severity, and nature of ocular and systemic adverse events (AEs) after multiple unilateral intravitreal (IVT) doses. **Measurement Method:** Clinical assessment including slit-lamp examination, indirect ophthalmology, and intraocular pressure (IOP) monitoring.

7.2 Secondary & Exploratory Endpoints 1. Pharmacokinetics (PK) and systemic exposure parameters of R07669330. 2. Pharmacodynamics (PD) and anti-drug antibody (ADA) immunogenicity incidence. 3. Efficacy signals as measured by Best Corrected Visual Acuity (BCVA) changes and anatomic alterations via Spectral Domain Optical Coherence Tomography (SD-OCT), Fundus Fluorescein Angiography (FFA), and Color Fundus Photography (CFP).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for systemic and ocular tolerability post-IVT injection via regular clinic visits. **Serious Adverse Events (SAEs):** Reported within 24 hours to the Sponsor and independent monitoring boards. **Data Monitoring Committee (DMC):** Utilized to review safety data between the open-label MAD Part 1A phase before escalating to randomized Parts 1B and 2.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive at least one dose of the investigational product). **Sample Size Justification:** $N = 132$, suitably powered to determine maximum tolerated dose (MTD) and safety profiling against active comparators. **Handling of Missing Data:** Appropriate imputation models applied per the sponsor's biostatistics guidelines for repeated-measure ophthalmic clinical trials.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite and remote monitoring to oversee investigational product handling and masked reading center assessments. **Audit Trail:** Validated systems for tracking electronic Case Report Forms (eCRF) and central reading center imaging data (SD-OCT/CFP/FFA).

11. Study Identifiers & Governance

Primary ID: BP45482 **Registry Number:** NCT06961370 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** A Study to Investigate Safety, Tolerability, Pharmacokinetics (PK), Pharmacodynamics (PD), and Immunogenicity of R07669330 in Participants With Geographic Atrophy (GA) Secondary to Age-related Macular Degeneration (AMD) **Official Regulatory Title:** A Phase I, Multi-center, Multi-part Study to Investigate Safety,

Tolerability, PK, PD, and Immunogenicity of R07669330
Intravitreal Injections in Participants With GA Secondary to AMD:
Part 1A: Open-label, MAD; Part 1B: Randomized PD Pilot; Part 2:
Masked, Randomized, Active-comparator-controlled.

12. Clinical Framework & Disease Ontologies

Primary Condition: Geographic Atrophy (GA) Secondary to Age-related Macular Degeneration (AMD) **Disease Areas:** Geographic Atrophy, Age-related Macular Degeneration **Condition Name:** Macular Degeneration **Preferred Name:** Macular degeneration (Preferred UMLS Name: Geographic atrophy) **Semantic Type:** Disease or Syndrome **Clinical Ontologies:** * **MeSH Code:** D008268 * **SNOMED ID:** 422338006 * **HPO Code:** HP:0000608 **Diagnosis Threshold:** Confirmed macular atrophy using SD-OCT and/or autofluorescence **Medical Methodology:** Direct Ocular Intervention (Intravitreal Injections) **Optimization Target:** Maintenance/stabilization of BCVA and reduction in GA lesion expansion

13. Study Procedures & Methodology

Management Pathway: Standard ophthalmologic pathway for AMD and GA **Education Protocol (HCP):** Masking procedures (for Part 1B and 2) and central imaging submission **Education Protocol (Patient):** Education regarding ocular injection procedures, expectations of multiple IVT doses, and reporting of visual changes **Quality Audit Mechanism:** Centralized reading center for objective grading of ophthalmic imaging (SD-OCT, FFA)

14. Observation & Follow-Up Schedule

Total Duration: Up to 18 months (Primary completion Jan 2027) **Onsite Visit Cadence:** Required for each scheduled unilateral IVT dose administration and safety assessment **Remote Monitoring Frequency:** Standard protocol intervals **Checklist Review Interval:** Prior to every injection (IOP checks, slit-lamp, indirect ophthalmology)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				